Comprehensive Analytical Questions

UK Rail Ticket Sales & Journey Performance (January 1 – April 30, 2024)
Structured Analysis and Scope Reference

1. Sales and Revenue

- What is the total revenue, number of transactions, and average ticket price?
- How does revenue trend by day, week, and month (January to April 2024)?
- Which months, weeks, and weekdays are the strongest and weakest?
- What is the month-over-month (MoM) and week-over-week (WoW) growth rate?
- Which routes generate the highest revenue, and which generate the lowest?
- What share of total revenue is contributed by the top 10 routes (Pareto / 80-20 analysis)?
- Do revenue and transaction volume move together, or do certain low-volume routes generate high revenue due to higher average pricing?
- How does average ticket pricing differ across various Railcard categories (Adult, Senior, Disabled, No Railcard) within the same route and ticket type?

2. Customer Purchasing Behavior

- What is the split between Online and Station purchases, and is it shifting over time?
- Which payment methods are most popular, and do they vary by purchase type or price band?
- At what hours of the day do customers purchase tickets (peak purchasing times)?
- How many days in advance do customers book their journeys (Date_of_Journey minus Date_of_Purchase)?
- Does lead time differ significantly by ticket type (Advance vs. Anytime vs. Off-Peak)?
- Do last-minute bookings for Advance tickets cost significantly more on average compared to early-bird bookings across different routes?
- Do purchase channels (Online vs. Station) differ in average ticket price, and do certain payment methods correlate with higher transaction values?

3. Ticket and Product Analysis

- Which ticket types, classes, and railcards sell the most, and which generate the highest revenue?
- What is the average price by ticket type, class, and route?
- Which railcard holders travel most frequently, and on which routes?
- What is the First Class vs. Standard Class mix, and does it vary across routes?
- Which route and ticket-type combinations perform best commercially?
- What is the revenue contribution share of First Class tickets compared to their actual transaction volume share?
- How does the distribution of ticket types (Advance, Anytime, Off-Peak) differ across various Railcard segments?

4. Route Performance

- Which routes and stations (departure and arrival) are the busiest?
- Which routes have the highest average price, and which appear most price-sensitive?
- Which stations act as the primary network hubs (highest departures and arrivals)?
- Are there popular return-trip pairs (A to B and B to A), and are their volumes balanced?
- Which routes show high volume but low revenue (underpriced) versus low revenue but high volume (premium)?
- The Disruption-Revenue Matrix: Which specific routes combine high passenger volume and revenue with high delay or cancellation rates (chronic bottlenecks)?
- Are reverse-direction route pairs balanced in terms of demand volume and average pricing?

5. Punctuality and Delays

- What is the overall on-time versus delayed percentage by transaction?
- What is the distribution of delay durations (Actual_Arrival_Time minus Arrival_Time) in terms of mean, median, and maximum?
- Which routes, departure stations, and arrival destinations experience the most delays?
- What are the most frequent reasons for delay (Reason_For_Delay), and which causes result in the longest delays?
- How do delays vary by hour of the day, weekday, month, and time band?
- Do peak-hour services experience longer delays than off-peak services?
- Is operational performance improving or worsening over time?
- How are delays distributed across severity buckets (0–5, 5–15, 15–30, and 30+ minutes)?
- Do specific routes suffer repeatedly from the same root cause (e.g., signal failure)?
- Normalized Disruption Metric: What is the total number of delay minutes per 100 transactions for each route to ensure a fair comparison?
- Do primary delay causes cluster heavily around specific hours of the day or days of the week?

6. Refunds and Customer Impact

- What is the overall refund request rate (Refund_Request = Yes)?
- Do delayed or cancelled journeys exhibit significantly higher refund request rates than on-time ones?
- How does the refund request rate change with delay length and delay reason?
- How much gross ticket revenue is associated with refund requests?
- Which routes, ticket types, and classes generate the highest volume of refund requests?
- Are Advance tickets requested for refunds more frequently than flexible tickets?
- Are refund requests submitted for journeys that were actually on time (indicating data anomalies or specific operational exceptions)?
- What is the exact breakdown of refund request rates mapped against each specific Reason_For_Delay category?